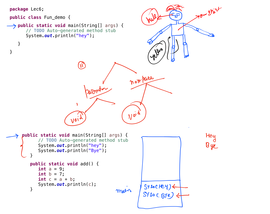
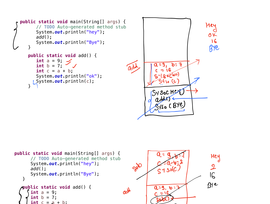
Lec-6
Monday, October

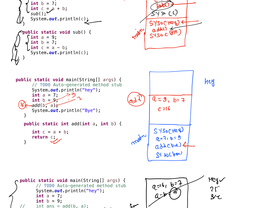
```
package Lec6;
public class Fun_demo {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("hey");
    }
}
```



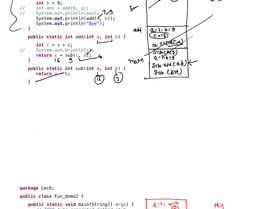
```
package Lec6;
public class Fun_demo {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("hey");
        System.out.print("bye");
    }
}
```



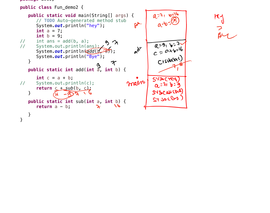
```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.print("hey");
    System.out.print("bye");

    public static void add() {
    int a = 5;
    int b = 7;
    int c = a + b;
    System.out.print(c);
    }
}
```



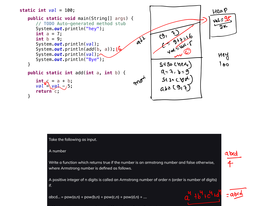
```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.print("hey");
    add();
    System.out.print("bye");

    public static void add() {
    int a = 5;
    int b = 7;
    int c = a + b;
    System.out.print("ca");
    System.out.print(c);
    }
}
```



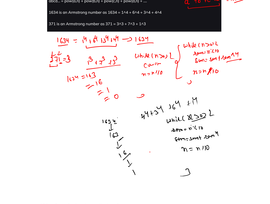
```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.print("hey");
    add();
    System.out.print("bye");

    public static void add() {
    int a = 5;
    int b = 7;
    int c = a + b;
    add();
    System.out.print(c);
    }

    public static void sub() {
    int a = 5;
    int b = 7;
    int c = a - b;
    System.out.print(c);
    }
}
```



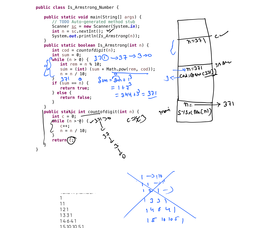
```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.print("hey");
    int a = 5;
    int b = 7;
    int c = a + b;
    add(a, b);
    System.out.print(c);
    System.out.print(add(a, b));
    System.out.print("bye");

    public static int add(int a, int b) {
    int c = a + b;
    return c;
    }
}
```



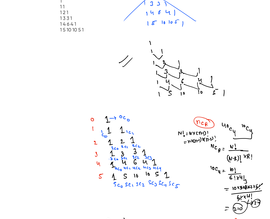
```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.print("hey");
    int a = 5;
    int b = 7;
    int c = a + b;
//  add(a, b);
//  System.out.print(c);
//  System.out.print(add(a, b));
    System.out.print("bye");

    public static int add(int a, int b) {
    int c = a + b;
    return c;
//  add(a, b);
    }

    public static int sub(int a, int b) {
    int c = a - b;
    return c;
    }
}
```



```
package Lec6;
public class Fun_demo2 {
    public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.print("hey");
    int a = 5;
    int b = 7;
    int c = a + b;
//  add(a, b);
//  System.out.print(c);
//  System.out.print(add(a, b));
    System.out.print("bye");

    public static int add(int a, int b) {
    int c = a + b;
//  add(a, b);
    return c + add(a, b);
    }

    public static int sub(int a, int b) {
    int c = a - b;
    return c - sub(a, b);
    }
}
}
```



```
static int val = 100;
    public static void main(String[] args) {
    // TODO Auto-generated method stub
    System.out.print("hey");
    int a = 5;
    int b = 7;
    int c = a + b;
    System.out.print(c);
    System.out.print(add(a, b));
    System.out.print("bye");

    public static int add(int a, int b) {
    int c = a + b;
    val = val - 5;
    return c;
    }
}
```



Take the following as input.

A number

Write a function which returns true if the number is an armstrong number and false otherwise.

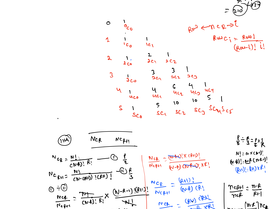
Lecture Armstrong number is defined as follows.

A positive integer of n digits is called an Armstrong number of order n (order is number of digits) if

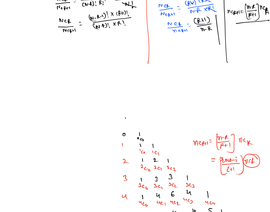
abcd... = $a^n + b^n + c^n + d^n + \ldots$

1634 is an Armstrong number as 1634 = $1^4 + 6^4 + 3^4 + 4^4$

371 is an Armstrong number as 371 = $3^3 + 7^3 + 1^3$



```
public class Is_Armstrong_Number {
    public static void main(String[] args) {
    // TODO Auto-generated method stub
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    System.out.print(Is_Armstrong(n));
    }
    public static boolean Is_Armstrong(int n) {
    int cnt = countOfDigits(n);
    int sum = 0;
    int temp = n;
    while (temp > 0) {
        int rem = n % 10;
        sum = (int) (sum + Math.pow(rem, cnt));
        temp = temp / 10;
    }
    if (sum == n) {
        return true;
    } else {
        return false;
    }
    }
    public static int countOfDigits(int n) {
    int cnt = 0;
    while (n > 0) {
        cnt = cnt + 1;
        n = n / 10;
    }
    return cnt;
    }
}
```



```
1
11
121
1331
14641
15101051
```